PHYS 143 – Mechanics and Thermal Physics

Potential Energy and Energy Conservation

What happens to energy when forces act along a path, and how can stored energy predict motion?

Hope Donglo

Office 24, Department of Physics, University of Ghana

Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh

Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

By the end of this week, you should be able to:

1. Calculate gravitational and elastic potential energy, and say which forces have a potential energy.
2. Apply conservation of mechanical energy to find speeds, heights and compressions.
3. Include friction and other forces in the energy equation, and say where the energy goes.
4. Find a force from a potential energy, and read an energy diagram.

Central question

What happens to energy when forces act along a path, and how can stored energy predict motion?

The thread of the week

work of some forces → stored energy U **→** $K + U$ **stays constant.** Other forces add or remove energy through their work: $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$.

Two contacts, 110 minutes each: A (Tuesday) **IDENTIFY** potential energy and conservation, then **QUANTIFY** springs, friction and other forces; **B** (Wednesday) **QUANTIFY** with the full energy equation on harder problems, then **EXTEND** to which forces have U and what $U(x)$ tells us.
Reused: Lecture 4 (work–energy, spring work), the dot product, free fall, free-body diagrams, circular motion.
New: U, E, conservative forces, $F_x = -\mathrm{d}U/\mathrm{d}x$, energy diagrams. **Later:** Lectures 6, 7, 8, 11.

Reading: Young and Freedman, Chapter 7, §7.1–7.5. **Labels:** **CORE** you should be able to do this; **PRACTICE** develops the skill; **REFERENCE** keep for later.

Not marked. Try each, then check.

1. A $2.0\,\mathrm{kg}$ mango falls $5.0\,\mathrm{m}$. Work done by gravity?
2. A $3.0\,\mathrm{kg}$ block slows from 4.0 to $2.0\,\mathrm{m/s}$. Net work?
3. A spring ($k = 200\,\mathrm{N/m}$) is stretched $0.10\,\mathrm{m}$. Work done on it?
4. A box slides $4.0\,\mathrm{m}$ down a slope against $12\,\mathrm{N}$ of friction. Work by friction?
5. A $0.50\,\mathrm{kg}$ stone is whirled at $4.0\,\mathrm{m/s}$ in a circle of radius $1.0\,\mathrm{m}$. Acceleration and net force?
6. Differentiate $y = 3x^2 - x^3$ at $x = 2$.

Answers

1. $mgh = (2.0)(9.8)(5.0) = 98\,\mathrm{J}$, positive: down, along the force.
2. $\Delta K = \frac{1}{2}(3.0)(2.0^2 - 4.0^2) = -18\,\mathrm{J}$
3. $\frac{1}{2}kx^2 = \frac{1}{2}(200)(0.10)^2 = 1.0\,\mathrm{J}$
4. $W_f = -fd = -(12)(4.0) = -48\,\mathrm{J}$
5. $a = v^2/r = 16\,\mathrm{m/s^2}$, toward the centre; $F = ma = 8.0\,\mathrm{N}$
6. $y' = 6x - 3x^2$; at $x = 2$: $12 - 12 = 0$

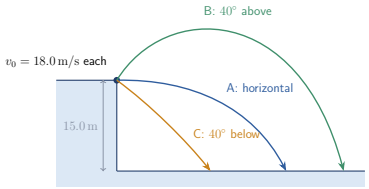
Items 1 to 4 are Lecture 4, item 5 is Week 3, item 6 is calculus. All come back this week; a derivative refresher is in the supplementary notes.

Contact A: Identify, then Quantify

What is potential energy, when is mechanical energy conserved, and how do springs and other forces enter the energy equation?

Lecture 4

$W_{\text{tot}} = \Delta K$. Work changes the speed. *Then what happens to it?*



Predict first

Mark the change in height of each stone on the sketch. Predict: A, B, C, or all equal?

Lecture 5

The work of some forces can be *stored* and given back. *Which forces, and how do we use that?*

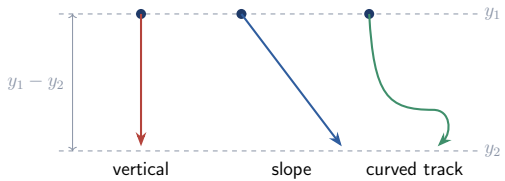
Opening question

Three students stand on the roof of a five-storey hostel, $15.0\,\text{m}$ above the ground. Each throws a stone at the same speed, $18.0\,\text{m/s}$: A horizontally, B at 40° above the horizontal, C at 40° below it. Ignore air resistance. Which stone hits the ground with the greatest speed?

The vocabulary of stored energy

Quantity	What it is	What it tells you	SI unit	How to picture it
Potential energy U	energy associated with the position of the parts of a system; the possibility for work to be done	the work that can still be done as the position changes, $W = -\Delta U$	joule, J	height of a bar in a bar chart; a curve $U(x)$
Mechanical energy E	the sum $E = K + U$ of the kinetic and potential energies of the system	all the energy of motion and position together	J	total height of the bars
System	the objects whose energies we add up: the object, the Earth, later the spring	what U belongs to		the box we draw round them
Reference level	the height where we choose $y = 0$	where $U_{\text{grav}} = 0$; a choice, not a fact	m	a line on the sketch

Reading the table. Kinetic energy asks: *how much work is stored in the motion?* Potential energy asks: *how much work could still be done as the position changes?* We build the idea one step at a time, starting with the work of gravity.



Key idea

Same start height, same end height: three different paths.

Predict first

A $2.0\,\mathrm{kg}$ bag goes from y_1 to y_2, $3.0\,\mathrm{m}$ lower, by each route. Is the work of gravity largest on the vertical, the slope, the curve, or the same on all three?

Rule: work done by gravity

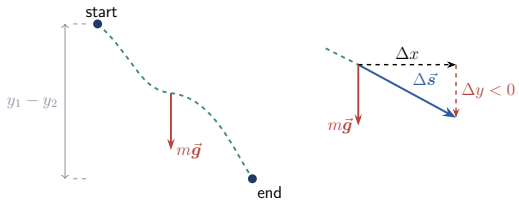
$$W_{\text{grav}} = mg\,(y_1 - y_2) \qquad (1)$$

Define. m is the mass (kg). $g = 9.80\,\mathrm{m/s^2}$, constant near the surface. y is the height above any level we choose, measured **upward**; y_1 is the start, y_2 the end (m). W_{grav} is a scalar in joules.

Explain. Equation (1) holds for *any* path from y_1 to y_2. Falling ($y_2 < y_1$): $W > 0$. Rising: $W < 0$. Same height at both ends: $W = 0$. *Reuse:* Lecture 4 (6)

But what if... you want to know why the shape of the path never enters? The next slide shows it step by step.

Why only the height counts



Key idea

A horizontal step does no work. Only the change Δy counts.

Explain. Cut the path into small steps $\Delta \vec{s} = \Delta x\, \hat{\imath} + \Delta y\, \hat{\jmath}$.

1. The weight is $\vec{w} = -mg\, \hat{\jmath}$ on every step.

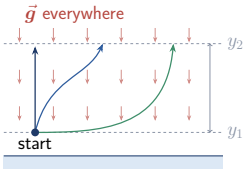
2. The dot product from Lecture 1 gives the work on one step: $\vec{w} \cdot \Delta \vec{s} = -mg\, \Delta y$. The Δx part drops out because $\hat{\jmath} \cdot \hat{\imath} = 0$.

3. Add all the steps. In the limit of small steps the sum is an integral: $W_{\text{grav}} = \int_{y_1}^{y_2} (-mg)\, \mathrm{d}y = mg\, y_1 - mg\, y_2 = mg\, (y_1 - y_2)$, whatever the route. This is (1).

Valid when g is constant: near the Earth's surface.

But what if... the work of gravity depends only on the two heights? Then the next frame puts it in a picture: a force field, and the work stored in it.

Gravity as a force field: where potential energy comes from



Key idea

Work done against a conservative force field is not lost; it is stored as potential energy due to the configuration of the system.

System, not object

The field and U belong to the *interaction* of the object with the Earth. U is a property of the configuration of the system (object plus Earth), not of the isolated object.

State. Near the surface the Earth sets up a uniform field $\vec{\boldsymbol{g}}$, $9.80\,\mathrm{m/s^2}$ downward, at every point. It acts on the object with no contact, whatever the path. The work *done by the field* from y_1 to y_2 is

$$W_{\text{field}} = \int \vec{\boldsymbol{F}} \cdot \mathrm{d}\vec{\boldsymbol{r}} = \int_{y_1}^{y_2} (-mg)\,\mathrm{d}y = -mg\,(y_2 - y_1),$$

which is (1) again. Raising the object ($y_2 > y_1$) at constant speed, *you* do $+mg\,(y_2 - y_1)$: the cost of lifting against the field.

Define. $\vec{\boldsymbol{F}} = m\vec{\boldsymbol{g}}$ is the weight (N); $\mathrm{d}\vec{\boldsymbol{r}}$ a small step along the path (m); $y_2 - y_1$ the height gained (m).

Explain. Only the height change enters, so the three routes cost the same. Nothing is lost: the field gives the work back when the object falls. That stored work is the **potential energy**, defined next.

Defining gravitational potential energy

Rule: gravitational potential energy

$$W_{\text{grav}} = -\Delta U_{\text{grav}} \tag{2}$$

$$U_{\text{grav}} = mgy \tag{3}$$

Define. U_{grav} is the gravitational potential energy (J), a scalar. $\Delta U_{\text{grav}} = U_2 - U_1$. y is the height above the **reference level** we choose, measured upward. m and g as in (1).

Explain. Compare (1) with (2): $mg(y_1 - y_2) = -(mgy_2 - mgy_1)$. Gravity gives back what the lift stored.

Valid near the surface, where g is constant.

	rising	falling
ΔU_{grav}	> 0	< 0
W_{grav}	< 0	> 0

Predict first

A $2.0\,\text{kg}$ bag is raised $3.0\,\text{m}$. Predict the sign of ΔU and of W_{grav}, then calculate.

$\Delta U = mg\Delta y = (2.0)(9.8)(3.0) = +59\,\text{J}$ and $W_{\text{grav}} = -59\,\text{J}$.

Common error

U_{grav} belongs to the object *and* the Earth together, not to the object alone.

The zero of U is a choice

A $0.50\,\mathrm{kg}$ mango falls from a branch $2.0\,\mathrm{m}$ above the ground onto a bench $0.50\,\mathrm{m}$ above the ground. Two students choose different reference levels.

Reference level ($y = 0$)	U_{branch}	U_{bench}	$\Delta U = U_{\text{bench}} - U_{\text{branch}}$
the ground	$+9.8\,\mathrm{J}$	$+2.45\,\mathrm{J}$	$-7.35\,\mathrm{J}$
$1.0\,\mathrm{m}$ above the ground ($y = +1.0$ and -0.5)	$+4.9\,\mathrm{J}$	$-2.45\,\mathrm{J}$	$-7.35\,\mathrm{J}$

Predict first

Which numbers change when the reference level moves, and which do not? Can U be negative?

What is physical

U itself changes with the reference level, and can be negative. The change ΔU, and so the work of gravity $W_{\text{grav}} = -\Delta U = +7.35\,\mathrm{J}$, is the same. Choose the level that makes the problem simplest (usually the lowest point), and keep it until the end.

Decide first, with a reason for each, then check

1. Can the gravitational potential energy of an object be negative?
2. Does U_{grav} belong to the object alone?
3. A ball is thrown upward. At the top its kinetic energy is zero. Where has that energy gone?
4. A ball falls with no air resistance. Is its kinetic energy conserved? Is its mechanical energy?

Answers

1. **Yes.** The zero is a choice; below the reference level $y < 0$.
2. **No.** It belongs to the object–Earth system.
3. **Into U_{grav}.** The ball has risen, so U has grown by the K that was lost.
4. K **no;** $E = K + U$ **yes.** K grows exactly as U falls.

But what if... K changes only by what U gives or takes? Prove it: combine the work–energy theorem with (2).

Rule: conservation of mechanical energy

$$K_1 + U_1 = K_2 + U_2 \qquad (4)$$

$E = K + U =$ constant

Define. $K = \frac{1}{2}mv^2$ is the kinetic energy (J). $U = U_{\text{grav}} = mgy$ for now. Subscripts 1 and 2 label two positions of the object. E is the total mechanical energy (J), a scalar.

Explain. Only gravity does work. Then:
1. Work–energy theorem: $W_{\text{tot}} = K_2 - K_1$. *Reuse:* Lecture 4 (5)
2. Only gravity works:
$W_{\text{tot}} = W_{\text{grav}} = -\Delta U = U_1 - U_2$, from (2).
3. Equate: $K_2 - K_1 = U_1 - U_2$, so
$K_1 + U_1 = K_2 + U_2$.

When it holds

Only conservative forces do work. Here that means gravity. A normal force or a tension acts perpendicular to the motion and does no work, so it may be present. If friction or a push does work, (4) fails: see (7), later in Contact A.

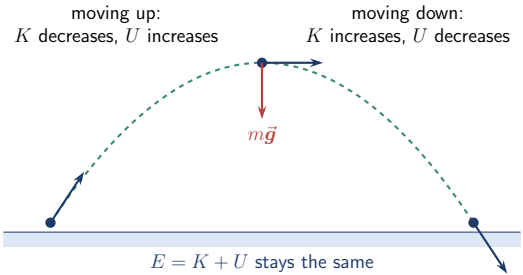
Predict first

A mango falls from rest at height h, no air resistance. Predict its speed at height $h/2$ before calculating: more or less than half its final speed?

At $h/2$, $\frac{1}{2}mv^2 = mg(h/2)$, so $v = \sqrt{gh}$, which is 0.71 of the final speed $\sqrt{2gh}$. More than half.

Energy changes form, not amount

PRACTICE



Explain. A long jumper rises, then falls. Going up, K is turned into U. Going down, U is turned back into K. At every point $E = K + U$ has the same value, if air resistance is small.

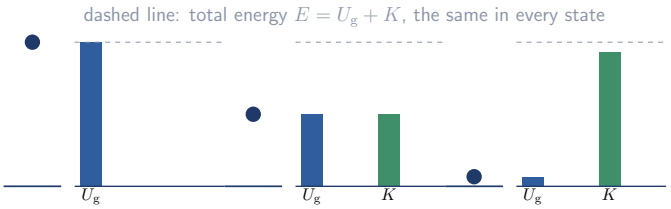
Common error

At the top of the jump K is *not* zero. The horizontal velocity is unchanged, so the speed there is at its smallest, not zero.

eg. a long jumper at an athletics meeting.

Bar charts: seeing K and U trade

CORE



Key idea

Draw a bar for each energy at each state. The totals must be equal. U_{el} and U_{int} join later.

Predict first

A mango falls from rest. Sketch the U and K bars at the halfway state before looking. What must their heights add to?

U falls and K rises by the same amount. K alone is not conserved. E is.

• METHOD – Using $K_1 + U_1 = K_2 + U_2$

1. STATES: choose position 1 and position 2. Mark the heights and the speeds on a sketch.
2. REFERENCE: choose $y = 0$, usually at the lowest point. Write it on the sketch.
3. FORCES: list every force on the object. Confirm that all except gravity do *no* work.
4. BARS: sketch K and U at each state. The totals must be equal.
5. WRITE $\frac{1}{2}mv_1^2 + mgy_1 = \frac{1}{2}mv_2^2 + mgy_2$ and solve. The mass often cancels.

WATCH OUT: use the vertical change in height, not the length of the slope; do not move the reference level halfway through.
CHECK: is the speed larger where the object is lower?

Which method?

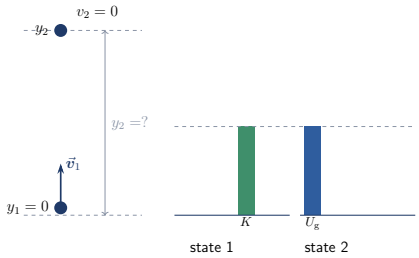
Energy (4): a *speed, height or distance* between two positions, when only gravity (and later a spring) does work. No time and no path shape needed.

Newton: an *acceleration, a time*, or a *force at one point*.

Decide: which method?

(a) Speed of a mango as it lands. (b) Time it takes to fall. (c) Maximum height of a thrown ball.

Answers: (a) energy, (b) Newton (or kinematics), (c) energy.



Key idea

All of K_1 becomes U_2 at the top.

Example 1

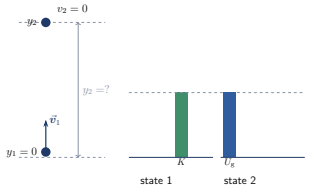
A boy on a school field throws a $0.145\,\mathrm{kg}$ ball straight up, giving it an initial speed of $20.0\,\mathrm{m/s}$. Find how high it goes. Ignore air resistance.

Predict first

Will the height depend on the mass? Is it nearer 2, 20 or $200\,\mathrm{m}$?

Example 1: a ball thrown straight up, solution

CORE



Key idea

All of K_1 becomes U_2 at the top.

IDENTIFY

Given: $m = 0.145\,\text{kg}$, $v_1 = 20.0\,\text{m/s}$, $v_2 = 0$ at the top. Find: the height y_2.

SET UP

Only gravity does work, so use (4). Reference $y = 0$ at the hand: $y_1 = 0$, $U_1 = 0$.

EXECUTE

Equation: $K_1 + U_1 = K_2 + U_2$, so $\frac{1}{2}mv_1^2 = mgy_2$

Substitute: $\frac{1}{2}(0.145\,\text{kg})(20.0\,\text{m/s})^2 = (0.145\,\text{kg})(9.80\,\text{m/s}^2)\,y_2$

Left side: $K_1 = 29.0\,\text{J}$

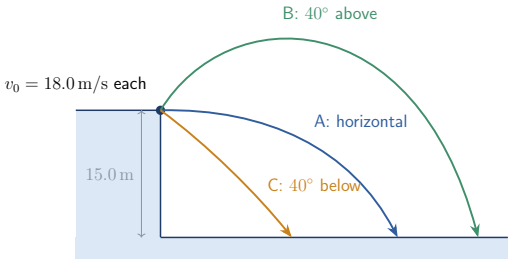
Solve: $y_2 = \dfrac{29.0\,\text{J}}{(0.145\,\text{kg})(9.80\,\text{m/s}^2)}$ $\qquad \boxed{y_2 = 20.4\,\text{m}}$

EVALUATE

Nearer $20\,\text{m}$, as predicted. The mass cancels: $y_2 = v_1^2/2g$, the Week 2 result. Units: $\text{J/N} = \text{m}$. ✓

Example 2: three stones from a hostel roof

CORE



Key idea

The three paths are very different. The heights are the same.

Example 2

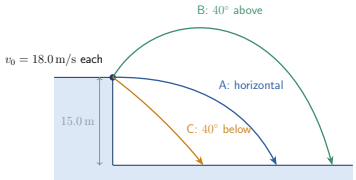
Three identical stones leave the roof of a $15.0\,\mathrm{m}$ hostel at the same speed, $18.0\,\mathrm{m/s}$: A horizontally, B at 40° above the horizontal, C at 40° below it. Ignore air resistance.
(a) Find the speed of each stone just before it hits the ground. Which is greatest?
(b) For stone B, find the greatest height above the roof and the speed there.

Predict first

Does the launch angle change the drop in height? Is the speed at the top of B's path zero?

Example 2: three stones, solution (a)

CORE



Key idea

The angle never appears in (4). Only v_0 and the drop do.

IDENTIFY

Given: $h = 15.0\,\mathrm{m}$, $v_0 = 18.0\,\mathrm{m/s}$ for A, B and C. Find: v at the ground.

SET UP

Only gravity does work (no air resistance), so use (4).
Reference $y = 0$ at the ground: $y_1 = 15.0\,\mathrm{m}$, $y_2 = 0$.

EXECUTE

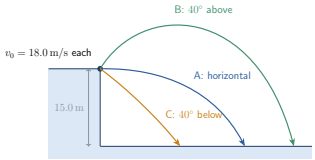
Equation: $\frac{1}{2}mv_0^2 + mgy_1 = \frac{1}{2}mv^2 + mgy_2$, so
$v = \sqrt{v_0^2 + 2g(y_1 - y_2)}$
Substitute: $v = \sqrt{(18.0\,\mathrm{m/s})^2 + 2(9.80\,\mathrm{m/s^2})(15.0\,\mathrm{m})}$
Add: $v = \sqrt{324 + 294}\,\mathrm{m/s} = \sqrt{618}\,\mathrm{m/s}$ $\boxed{v = 24.9\,\mathrm{m/s}}$

EVALUATE

The same for A, B and C: no stone is fastest. Between 18 and $30\,\mathrm{m/s}$, as expected. The angle changes the time in the air and where the stone lands, not the speed on landing. ✓

Example 2: three stones, solution (b)

CORE



Key idea

At the top $v_y = 0$ but v_x is unchanged: no horizontal force.

IDENTIFY

Given: B leaves at $v_0 = 18.0\,\mathrm{m/s}$, $40°$ above the horizontal. Find: the greatest height H above the roof, and the speed there.

SET UP

State 1: launch ($y = 0$ at the roof). State 2: highest point, where $v_x = v_0 \cos 40°$ and $v_y = 0$. Use (4).

EXECUTE

Equation: $\frac{1}{2}mv_0^2 = \frac{1}{2}m(v_0 \cos 40°)^2 + mgH$, so $H = v_0^2 \sin^2 40°/2g$

Substitute:

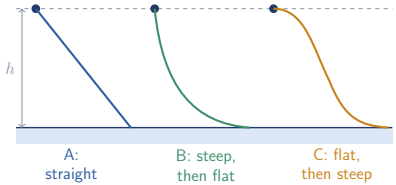
$H = (18.0\,\mathrm{m/s})^2(0.6428)^2/[2(9.80\,\mathrm{m/s^2})] = (324)(0.4132)/19.6\,\mathrm{m}$

$\boxed{H = 6.83\,\mathrm{m}}$

Speed at the top: $v_0 \cos 40° = (18.0)(0.7660)\,\mathrm{m/s}$ $\boxed{13.8\,\mathrm{m/s}}$, not zero

EVALUATE

Matches the Week 2 result $H = (v_0 \sin\theta)^2/2g$. ✓Likewise, stones thrown at the same speed at different angles have equal speeds at any common height: (4) contains only v_0 and the height.



Three frictionless tracks start at the same height h and end at the same level. A block is released from rest at the top of each.

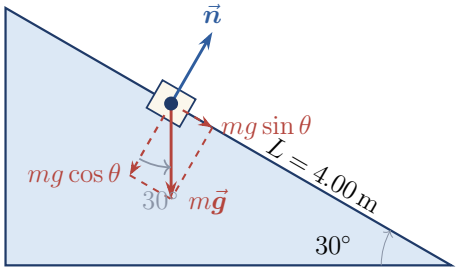
Key idea

Same height, same end level. Different shapes.

Answer, with a reason

(a) Compare the speeds at the bottom. (b) Must the arrival times be equal? (c) If the reference level moves to the top, which of U_1, U_2, ΔU change?

(a) All equal: $v = \sqrt{2gh}$, because the normal force does no work. **(b)** No: the acceleration along each track differs (Week 3), so the times differ. **(c)** U_1 and U_2 change; ΔU does not.



Key idea

The normal force does no work, so it never appears in the energy equation.

Example 3

A $20.0\,\mathrm{kg}$ sack of cement slides from rest down a smooth plank $4.00\,\mathrm{m}$ long, inclined at $30.0°$. Find (a) the normal force on the sack, (b) its acceleration, and (c) its speed at the bottom, by Newton's second law and again by energy.

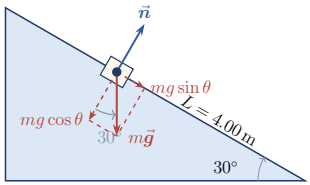
Predict first

Is n equal to mg? Do the two methods give the same speed?

Hint for the Newton route: the speed from constant acceleration, $v^2 = v_0^2 + 2aL$, with $a = g\sin\theta$ (Week 2).

Example 3: a smooth plank, by energy and by Newton, solution

PRACTICE



Key idea

The normal force does no work, so it never appears in the energy equation.

IDENTIFY

Given: $m = 20.0\,\mathrm{kg}$, $\theta = 30.0°$, $L = 4.00\,\mathrm{m}$, $v_1 = 0$. Find: n, a, v_2.

EXECUTE: Newton (Week 3)

Perpendicular to the plank: $n - mg\cos\theta = 0$, so

$n = (20.0)(9.80)(0.8660)\,\mathrm{N}$ $\boxed{n = 170\,\mathrm{N}}$

Along the plank: $mg\sin\theta = ma$, so $a = g\sin\theta = (9.80)(0.500)$

$\boxed{a = 4.90\,\mathrm{m/s^2}}$

Speed: $v^2 = v_0^2 + 2aL = 0 + 2(4.90)(4.00) = 39.2$ $\boxed{v = 6.26\,\mathrm{m/s}}$

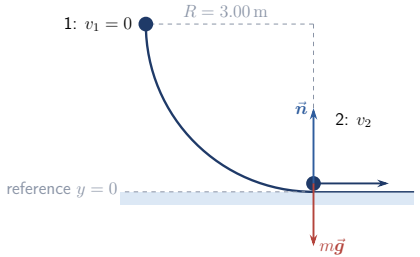
EXECUTE: energy (4)

Drop: $h = L\sin\theta = 2.00\,\mathrm{m}$. Equation: $mgh = \frac{1}{2}mv^2$, so

$v = \sqrt{2gh} = \sqrt{2(9.80)(2.00)}$ $\boxed{6.26\,\mathrm{m/s}}$

EVALUATE

The methods agree. Energy gave v in one step and never used n; Newton gave n and a too. ✓



Key idea

The normal force changes direction along the path but is always perpendicular to the motion.

Example 4

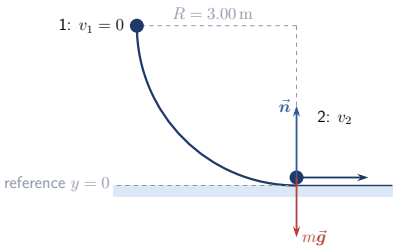
Kofi has a mass of $25.0\,\mathrm{kg}$ with his skateboard. He rolls down a smooth curved ramp that is a quarter of a circle of radius $R = 3.00\,\mathrm{m}$, starting from rest. Treat him as a particle. Find (a) his speed at the bottom, and (b) the normal force of the ramp on him there.

Predict first

Sketch the free-body diagram at the bottom. Is n less than, equal to, or greater than his weight, $245\,\mathrm{N}$?

Example 4: a playground ramp, solution (a): energy for the speed

CORE



Key idea

$n \perp \vec{s}$ at every point, so only gravity does work.

IDENTIFY

Given: $m = 25.0\,\mathrm{kg}$, $R = 3.00\,\mathrm{m}$, $v_1 = 0$. Find: v_2 at the bottom.

SET UP

n is perpendicular to the motion, so only gravity does work: use (4). Reference $y = 0$ at the bottom: $y_1 = R$, $y_2 = 0$.

EXECUTE

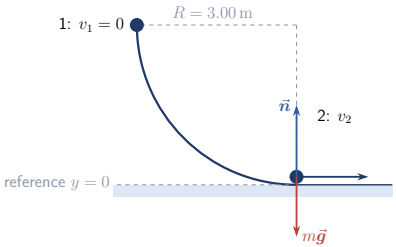
Equation: $K_1 + U_1 = K_2 + U_2$, so
$0 + mgR = \frac{1}{2}mv_2^2 + 0$, giving $v_2 = \sqrt{2gR}$
Substitute: $v_2 = \sqrt{2(9.80\,\mathrm{m/s^2})(3.00\,\mathrm{m})}$
Solve: $v_2 = \sqrt{58.8}\,\mathrm{m/s}$ $\boxed{v_2 = 7.67\,\mathrm{m/s}}$

EVALUATE

The mass cancels. The speed is what he would have after falling $3.00\,\mathrm{m}$ straight down: energy ignores the shape of the path. ✓

Example 4: a playground ramp, solution (b): Newton for the force

CORE



Key idea

Energy gave v. A force at one point needs $\sum \vec{F} = m\vec{a}$.

IDENTIFY

Given: $m = 25.0\,\mathrm{kg}$, $R = 3.00\,\mathrm{m}$, $v_2^2 = 2gR$ from (a).
Find: n at the bottom.

SET UP

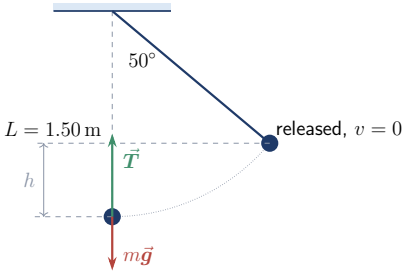
At the bottom the path is a circle with its centre above him, so the acceleration is $a_{\mathrm{rad}} = v_2^2/R$, upward. Take up as positive. *Reuse:* Lecture 3 (10), circular motion

EXECUTE

Equation: $\sum F_y = n - mg = ma_{\mathrm{rad}} = mv_2^2/R$
Substitute $v_2^2 = 2gR$: $a_{\mathrm{rad}} = 2gR/R = 2g$, so
$n = mg + 2mg = 3mg$
Solve: $n = 3(25.0\,\mathrm{kg})(9.80\,\mathrm{m/s^2})$ $\boxed{n = 735\,\mathrm{N}}$

EVALUATE

$n = 735\,\mathrm{N} > mg = 245\,\mathrm{N}$, as predicted: he accelerates upward, toward the centre. $n = 3mg$ for any radius. Energy gave the speed, Newton the force. ✓



Key idea

The height drop is $L(1 - \cos\theta)$, not L.

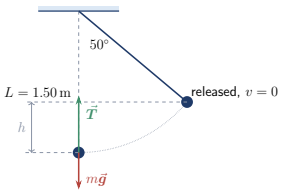
Independent problem

A $0.80\,\mathrm{kg}$ bob hangs on a string of length $L = 1.50\,\mathrm{m}$. It is released from rest with the string 50° from the vertical. Find (a) its speed at the lowest point, and (b) the tension in the string there.

Predict first

Sketch the free-body diagram at the bottom. Is T greater than, equal to, or less than $mg = 7.84\,\mathrm{N}$?

Your turn: a pendulum, solution



Key idea

The height drop is $L(1 - \cos\theta)$, not L.

IDENTIFY

Given: $m = 0.80\,\text{kg}$, $L = 1.50\,\text{m}$, $\theta = 50°$, $v_1 = 0$. Find v_2 and T.

SET UP

T is perpendicular to the path, so use (4), reference at the lowest point: $h = L(1 - \cos 50°) = 1.50(1 - 0.6428) = 0.536\,\text{m}$. Then Newton at the bottom.

EXECUTE

(a) Equation: $mgh = \frac{1}{2}mv^2$, so $v = \sqrt{2gh} = \sqrt{2(9.80)(0.536)}$

$$\boxed{v = 3.24\,\text{m/s}}$$

(b) Equation: $T - mg = mv^2/L$, so $T = mg + mv^2/L$

Substitute: $T = 7.84\,\text{N} + \dfrac{(0.80\,\text{kg})(10.5\,\text{m}^2/\text{s}^2)}{1.50\,\text{m}} = 7.84\,\text{N} + 5.60\,\text{N}$

$$\boxed{T = 13.4\,\text{N}}$$

EVALUATE

$T > mg$, as predicted. Check: $T = mg(3 - 2\cos 50°) = 13.4\,\text{N}$. ✓

Common mistakes and the correct habit

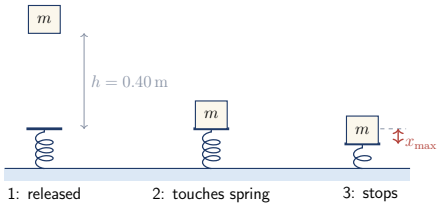
Mistake	Instead
Moving the reference level halfway through a problem.	Choose it once. Only ΔU is physical.
Using the length of the slope in $U_{\text{grav}} = mgy$.	Use the vertical height above the reference level.
Using (4) when friction or a push does work.	Add the work of those forces: rule (7), later in Contact A.
Counting gravity twice, in U and as a work term.	Gravity is already in U_{grav}.
Saying the kinetic energy is conserved.	The sum $E = K + U$ is conserved, not K.
Asking energy for a force or a time.	Energy gives speeds; use Newton for a force at a point.

But what if... a real ramp has friction, so Kofi reaches the bottom more slowly than $7.67\,\text{m/s}$? Where does the rest of the energy go? A spring raises a second question: how much energy does it store? Both come next.

A new question: springs and friction

So far: the work of gravity is stored as U_{grav}, and $K + U$ stays constant when only gravity does work.

Not yet answered: a spring stores energy too: how much? A rough ramp slows Kofi: where does his energy go?



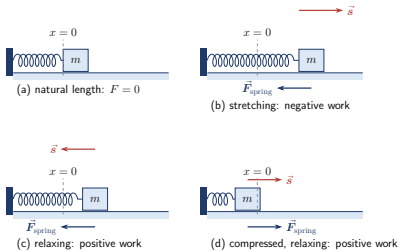
Poll

A block is dropped from $0.40\,\text{m}$ above a light vertical spring. At state 3 it stops for an instant. Is state 3 the position where the spring force equals the block's weight (the equilibrium point)?

Predict first

Sketch the speed against height. Is the block fastest at the equilibrium point, or at the instant it touches the spring?

Work done by a spring



Key idea

The spring force points toward $x = 0$: positive work moving toward it, negative moving away.

Predict first

A spring relaxes from $x_1 = 0.10\,\text{m}$ to $x_2 = 0.04\,\text{m}$. Predict the sign of W_{el}.

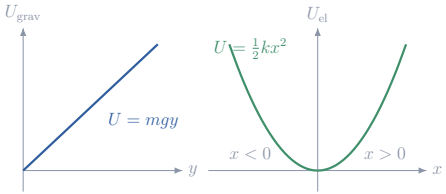
Rule: work done by a spring

$$W_{\text{el}} = \tfrac{1}{2}kx_1^2 - \tfrac{1}{2}kx_2^2 \qquad (5)$$

Define. W_{el} is the work done *by* the spring on the object (J). k is the force constant (N/m). x is the extension from the **natural length**: $x > 0$ stretched, $x < 0$ compressed (m). x_1 is the start, x_2 the end. *Reuse:* Lecture 4 (9)

Explain. Hooke's law gives $F_x = -kx$ on the block, so the force varies. The work is the integral $W_{\text{el}} = \int_{x_1}^{x_2}(-kx)\,\mathrm{d}x = \tfrac{1}{2}kx_1^2 - \tfrac{1}{2}kx_2^2$, the area under the force graph: this is (5). Stretching, $|x|$ grows and $W_{\text{el}} < 0$ (b). Relaxing from either side, $|x|$ shrinks and $W_{\text{el}} > 0$ (c), (d).

Valid for an ideal (light) spring within its elastic range.



straight line; zero is a choice

parabola; zero at natural length

Key idea

Gravity: a straight line, the zero is a choice. Spring: a parabola, the zero is the natural length.

Rule: elastic potential energy

$$W_{\text{el}} = -\Delta U_{\text{el}} \qquad U_{\text{el}} = \tfrac{1}{2}kx^2 \quad (6)$$

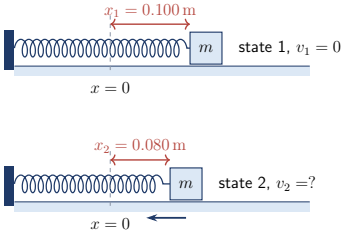
Define. U_{el} is the energy stored in the spring (J), a scalar; $\Delta U_{\text{el}} = U_{\text{el},2} - U_{\text{el},1}$. k, x as in (5).

Explain. (5) is $-(U_{\text{el},2} - U_{\text{el},1})$ with $U_{\text{el}} = \frac{1}{2}kx^2$. $U_{\text{el}} \geq 0$ for stretch or compression; doubling x quadruples it. It is stored in the spring itself, not shared with the Earth. **Valid** within Hooke's law, for a light spring.

Predict first

A spring is compressed by x, then by $3x$. Predict the ratio of the stored energies.

$U_{\text{el}} \propto x^2$, so the ratio is 9.



Key idea

Measure x from the natural length, $x = 0$, not from the wall.

Independent problem

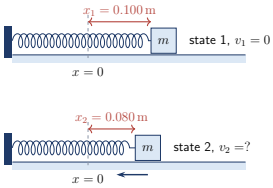
In the lab a $0.200\,\mathrm{kg}$ glider sits on a frictionless horizontal air track, joined to a spring with $k = 5.00\,\mathrm{N/m}$. You pull the glider so the spring stretches $0.100\,\mathrm{m}$ and release it from rest. What is its speed when the stretch is $0.080\,\mathrm{m}$?

Predict first

Less or more than the speed at $x = 0$, which is $x_1\sqrt{k/m} = 0.50\,\mathrm{m/s}$?

Your turn: a glider on an air track, solution

CORE



Key idea

Measure x from the natural length, $x = 0$, not from the wall.

IDENTIFY

Given: $m = 0.200\,\text{kg}$, $k = 5.00\,\text{N/m}$, $x_1 = 0.100\,\text{m}$, $v_1 = 0$, $x_2 = 0.080\,\text{m}$. Find v_2.

SET UP

n and w are perpendicular to the motion and the track is frictionless: only the spring does work. Use (4) with $U = U_{\text{el}}$.

EXECUTE

Equation: $\frac{1}{2}mv_1^2 + \frac{1}{2}kx_1^2 = \frac{1}{2}mv_2^2 + \frac{1}{2}kx_2^2$

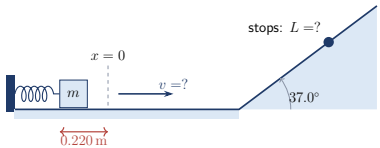
Energies: $U_1 = \frac{1}{2}(5.00)(0.100)^2 = 0.0250\,\text{J}$,
$U_2 = \frac{1}{2}(5.00)(0.080)^2 = 0.0160\,\text{J}$

Solve: $K_2 = 0 + 0.0250 - 0.0160 = 0.0090\,\text{J}$, so $v_2 = \sqrt{\dfrac{2(0.0090\,\text{J})}{0.200\,\text{kg}}}$

$$\boxed{v_2 = 0.30\,\text{m/s}}$$

EVALUATE

Less than $0.50\,\text{m/s}$, as predicted. The glider moves toward $x = 0$, so $v_x = -0.30\,\text{m/s}$. ✓



Key idea

Two stages, one equation: the spring's energy becomes K on the level, then U_{grav} on the ramp.

Independent problem

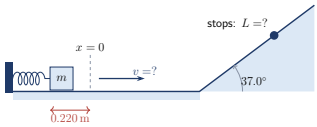
A $2.00\,\mathrm{kg}$ block is pushed against a light spring ($k = 400\,\mathrm{N/m}$), compressing it $0.220\,\mathrm{m}$. Released, it moves along a frictionless horizontal surface and then up a frictionless ramp inclined at $37.0°$. (a) How fast is it moving on the level after leaving the spring? (b) How far up the ramp does it go before sliding back?

Predict first

The stored energy is about $10\,\mathrm{J}$. A few m/s on the level? Less than a metre up the ramp?

Your turn: a spring launches a block up a smooth ramp, solution

PRACTICE



Key idea

Two stages, one equation: the spring's energy becomes K on the level, then U_{grav} on the ramp.

IDENTIFY

Given: $m = 2.00\,\text{kg}$, $k = 400\,\text{N/m}$, $x_1 = 0.220\,\text{m}$, $\theta = 37.0°$, no friction.
Find: v on the level; the distance L along the ramp.

SET UP

Only the spring and gravity do work: use (4) with $U = U_{\text{el}} + U_{\text{grav}}$.
Reference $y = 0$ on the level.

EXECUTE (a)

Equation: $\frac{1}{2}kx_1^2 = \frac{1}{2}mv^2$, so $v = x_1\sqrt{k/m}$

Substitute: $v = (0.220\,\text{m})\sqrt{(400\,\text{N/m})/(2.00\,\text{kg})}$ $\boxed{v = 3.11\,\text{m/s}}$

EXECUTE (b)

Equation: $\frac{1}{2}kx_1^2 = mgL\sin\theta$, so $L = kx_1^2/(2mg\sin\theta)$

Substitute: $L = (400)(0.220)^2/[2(2.00)(9.80)\sin 37.0°] = 19.36/23.59\,\text{m}$

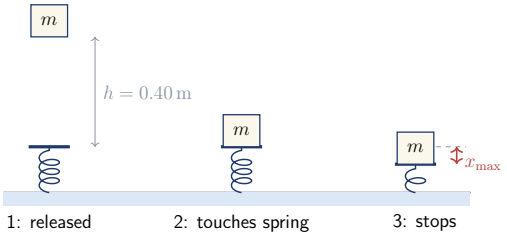
$\boxed{L = 0.821\,\text{m}}$

EVALUATE

A few m/s and under a metre, as predicted. Kinematic check:
$a = g\sin 37° = 5.90\,\text{m/s}^2$ and $v^2/2a = 0.821\,\text{m}$. ✓

Example 5: a block dropped on a spring

CORE



Key idea

Two potential energies, one reference level: $y = 0$ at the lowest point.

Example 5

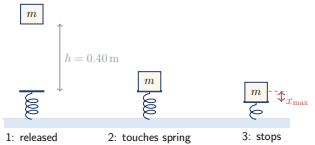
A $2.00\,\mathrm{kg}$ block is released from rest $0.40\,\mathrm{m}$ above a light vertical spring ($k = 1500\,\mathrm{N/m}$). It stops for an instant at the greatest compression, state 3.
(a) Find the greatest compression $x_{\max}$.
(b) Answer the poll: is state 3 the equilibrium point?
(c) Where is the block moving fastest?

Predict first

Estimate $x_{\max}$ from $\frac{1}{2}kx^2 = mgh$. Will the true value be larger or smaller? Compare with the equilibrium compression mg/k.

Example 5: a block dropped on a spring, solution (a)

CORE



Key idea

The block also falls x_{max} while the spring compresses: include it.

IDENTIFY

Given: $m = 2.00\,\text{kg}$, $h = 0.40\,\text{m}$, $k = 1500\,\text{N/m}$, $v_1 = v_3 = 0$. Find: x_{max}.

SET UP

Gravity and the spring do all the work: use (4) with $U = U_{grav} + U_{el}$. Reference $y = 0$ at state 3. State 1: $y_1 = h + x_{max}$, spring at natural length. State 3: $y_3 = 0$, $x = x_{max}$.

EXECUTE

Equation: $mg(h + x_{max}) = \frac{1}{2}kx_{max}^2$, a quadratic in x_{max}

Substitute: $(2.00)(9.80)(0.40 + x) = \frac{1}{2}(1500)x^2$, so

$750x^2 - 19.6x - 7.84 = 0$

Solve:

$x_{max} = \left[19.6 + \sqrt{19.6^2 + 4(750)(7.84)}\right]/2(750) = (19.6 + 154.6)/1500$

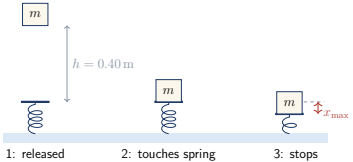
$\boxed{x_{max} = 0.116\,\text{m}}$

EVALUATE

The rough estimate $\sqrt{2mgh/k} = 0.102\,\text{m}$ is smaller, as predicted: it leaves out the energy mgx_{max} released during the compression. ✓

Example 5: the poll answered, and where the block is fastest

CORE



Key idea

Fastest where the net force is zero, not where the block stops.

(b) The poll: no

Equilibrium compression: $mg/k = 19.6/1500 = 0.013\,\text{m}$.
It stops at $x_{\max} = 0.116\,\text{m}$, where $kx_{\max} = 174\,\text{N}$, about nine times the weight ($19.6\,\text{N}$). The net force is upward, so the block turns back.

(c) Where is it fastest?

At $x = mg/k = 0.013\,\text{m}$, where the net force is zero. Until there, the weight is greater than the spring force and the block still speeds up. Beyond it the spring slows the block.
Numbers: at first contact $v = \sqrt{2gh} = 2.80\,\text{m/s}$. At $x = 0.013\,\text{m}$:
$\frac{1}{2}mv^2 = mg(h + x) - \frac{1}{2}kx^2 = 8.10 - 0.13 = 7.97\,\text{J}$, so $v = 2.82\,\text{m/s}$.

EVALUATE

The block stops far below the equilibrium point because it arrives with kinetic energy. It overshoots, and so a hanging mass oscillates. ✓

Rule: energy equation with other forces

$$K_1 + U_1 + W_{\text{other}} = K_2 + U_2 \qquad (7)$$

$$U = U_{\text{grav}} + U_{\text{el}}, \quad W_{\text{other}} = \Delta E = \Delta(K + U)$$

Define. W_{other} is the work (J) done by all forces **other than** gravity and the spring: friction, a push, a motor, air drag. It is a signed scalar. K, U as before; $E = K + U$.

Explain. The theorem gives $W_{\text{grav}} + W_{\text{el}} + W_{\text{other}} = \Delta K$. Replace the first two by $-\Delta U$, from (2) and (6), and rearrange: (7). Gravity and the spring are *inside* U, so never list them again as work.

Valid for any forces, once each is placed in U or in W_{other}.

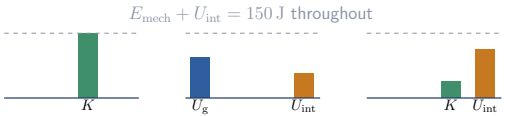
$W_{\text{other}} > 0$	adds: a push, a motor
$W_{\text{other}} < 0$	removes: friction, drag
$W_{\text{other}} = 0$	E constant: Equation (4)

Predict first

A box slides down a rough slope. Predict the sign of W_{other}, and whether its final speed is larger or smaller than on a smooth slope.

But what if... friction takes energy out of the mechanical bars? Where does it go?

Friction, and where the lost energy goes



1: bottom
$K = 150\,\text{J}$

2: highest point
$U_g = 94\,\text{J}$

3: back at the bottom
$K = 38\,\text{J}$

Key idea

$K + U$ falls from 150 to 38 J; the orange bar grows to match.

Rule: kinetic friction

$$W_f = -f_k d \tag{8a}$$

$$\Delta U_{\text{int}} = +f_k d \tag{8b}$$

Define. $f_k = \mu_k n$ is the magnitude of the kinetic friction force (N). d is the *path length* slid (m). U_{int} is the internal energy: energy associated with the state of the materials, for example hotter surfaces (J).

Explain. Friction points opposite to the motion at every instant, so each step contributes $-f_k\,\Delta s$ and the total is $-f_k d$, even if the path turns back. The mechanical energy removed becomes internal energy of the block and the surface: both warm up. It is not destroyed.

Valid for sliding friction of constant magnitude.

eg. the hot brake drums of a trotro coming down the Aburi hills.

Common mistakes and the correct habit

Mistake	Instead
Measuring the spring's stretch from the wall or from the equilibrium point.	From the natural length.
Taking the equilibrium position as where a dropped block stops.	It overshoots. Find the greatest compression from energy.
Leaving out U_{el} or U_{grav} for a vertical spring.	Include both, with one reference level.
Using friction's displacement instead of its path length.	$W_f = -f_k d$, with d the path length.
Listing gravity or the spring in W_{other} as well as in U.	They are in U already; W_{other} is the rest.
Saying friction "uses up" the energy.	It becomes internal energy: $\Delta U_{\text{int}} = f_k d$.

Contact A: what you can now do

- Calculate $U_{\text{grav}} = mgy$ and $U_{\text{el}} = \frac{1}{2}kx^2$, and say why only ΔU matters.
- Use $K_1 + U_1 = K_2 + U_2$ when only gravity and a spring do work, and choose between energy and Newton.
- Write $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$ and say where the energy lost to friction goes.

Predict first

Before Wednesday, predict: a crate slides up a rough ramp, stops and slides back. Does it return to the bottom with the speed it started with? Gravity and a spring can store energy: why can friction not? And if you know a formula for $U(x)$, can you find the force?

But what if... real problems combine springs, friction and slopes, and some forces cannot be stored at all? $\rightarrow$ **the full energy equation on harder problems, which forces have a potential energy, and what $U(x)$ tells us.**

Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh

Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh

Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

Contact B: Quantify, then Extend

How do we use the full energy equation on harder problems, which forces have a potential energy, and what does $U(x)$ tell us?

Predict first

1. A $0.50\,\mathrm{kg}$ ball falls from rest through $5.0\,\mathrm{m}$, with no air resistance. Find its speed at the bottom. $K_1 + U_1 = K_2 + U_2$ gives $v = \sqrt{2gh} = \sqrt{2(9.8)(5.0)} = 9.9\,\mathrm{m/s}$. The mass cancels.
2. A spring with $k = 400\,\mathrm{N/m}$ is compressed $0.050\,\mathrm{m}$. How much energy does it store? $U_{\mathrm{el}} = \frac{1}{2}kx^2 = \frac{1}{2}(400)(0.050)^2 = 0.50\,\mathrm{J}$, measured from the natural length.
3. A $3.0\,\mathrm{kg}$ block slides $2.0\,\mathrm{m}$ along a rough floor against a $6.0\,\mathrm{N}$ friction force. Find W_f and ΔU_{int}. $W_f = -f_k d = -(6.0)(2.0) = -12\,\mathrm{J}$, so $\Delta U_{\mathrm{int}} = +12\,\mathrm{J}$: the energy becomes internal energy.

• METHOD – Energy with other forces

1. SORT THE FORCES: gravity $\rightarrow U_{\text{grav}}$; spring $\rightarrow U_{\text{el}}$; the rest $\rightarrow W_{\text{other}}$ (perpendicular forces give zero).
2. STATES: sketch positions 1 and 2 with heights, speeds and spring extensions. One reference level.
3. W_{other}**:** sign first (friction < 0), then size ($f_k d$, path length).
4. BARS: K, U_{grav}, U_{el}, and U_{int} if friction acts.
5. WRITE (7) and solve.

WATCH OUT: gravity or the spring counted again in W_{other}; the displacement used for friction.
CHECK: slower than the frictionless case? A second method agrees?

Which method?

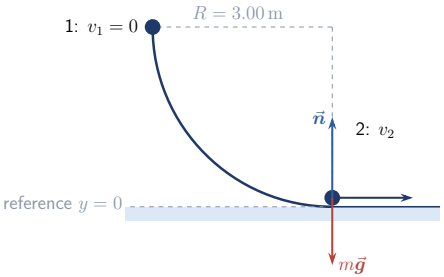
Energy (7): a speed, height, distance, compression or friction force between two states.

Newton: an acceleration, a time, or a force at one point.

Decide

(a) Compression of a lift's spring. (b) Time for a sack to slide down a rough plank. (c) Friction force from a stopping distance.

Answers: (a) energy, (b) Newton, (c) energy.



Key idea

Same states as Example 4. Now W_{other} is the unknown.

Example 6

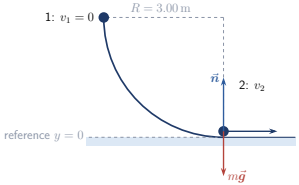
The ramp of Example 4 is rough. Kofi ($25.0\,\mathrm{kg}$, with his board) starts from rest at the top and reaches the bottom at $6.00\,\mathrm{m/s}$. How much work does friction do on him?

Predict first

Positive or negative? Is it less than the $735\,\mathrm{J}$ he started with?

Example 6: Kofi on a rough ramp, solution

CORE



Key idea

Same states as Example 4.
Now W_{other} is the unknown.

IDENTIFY

Given: $m = 25.0\,\mathrm{kg}$, $R = 3.00\,\mathrm{m}$, $v_1 = 0$, $v_2 = 6.00\,\mathrm{m/s}$. Find: W_f.

SET UP

Gravity is in U. $n \perp \vec{s}$: zero. Friction is the only other force, so $W_f = W_{\text{other}}$. Use (7), reference at the bottom: $y_1 = R$, $y_2 = 0$.

EXECUTE

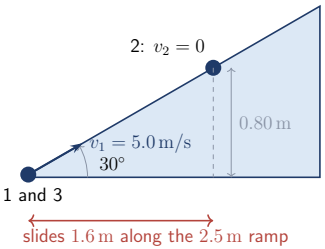
Equation: $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$, so $W_{\text{other}} = K_2 + U_2 - K_1 - U_1$
Substitute: $W_f = \frac{1}{2}(25.0)(6.00)^2 + 0 - 0 - (25.0)(9.80)(3.00)$
Solve: $W_f = 450\,\mathrm{J} - 735\,\mathrm{J}$ $\boxed{W_f = -285\,\mathrm{J}}$

EVALUATE

Negative and smaller in size than $735\,\mathrm{J}$, as predicted. His speed, $6.00\,\mathrm{m/s}$, is below the frictionless $7.67\,\mathrm{m/s}$. About 39% of the starting energy became internal energy. ✓

Example 7: a crate of tomatoes up a ramp

CORE



Key idea

Three states: bottom, highest point, bottom again. Friction acts over the whole path, up and back.

Example 7

A trader loads a $12.0\,\mathrm{kg}$ crate of tomatoes onto a pickup truck by sliding it up a ramp $2.5\,\mathrm{m}$ long, inclined at $30°$. She gives it $5.0\,\mathrm{m/s}$ at the bottom and lets go. Friction is not negligible: the crate slides $1.6\,\mathrm{m}$ up, stops, and slides back down.

(a) Taking the friction force as constant, find its size.

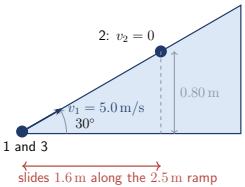
(b) How fast is the crate moving when it reaches the bottom again?

Predict first

Will the speed on return be more or less than $5.0\,\mathrm{m/s}$?

Example 7: a crate of tomatoes up a ramp, solution

CORE



Key idea

d is the path: 1.6 m up, then 3.2 m for the round trip.

IDENTIFY

Given: $m = 12.0\,\mathrm{kg}$, $v_1 = 5.0\,\mathrm{m/s}$, $d = 1.6\,\mathrm{m}$ up the 30° ramp, $v_2 = 0$ at the top. Find: f and v_3.

SET UP

Reference $y = 0$ at the bottom. Top: $y_2 = 1.6 \sin 30° = 0.80\,\mathrm{m}$. n does no work; friction is W_{other}. Use (7).

EXECUTE: (a) states 1 to 2

Equation: $\frac{1}{2}mv_1^2 + 0 - fd = 0 + mgy_2$

Substitute: $\frac{1}{2}(12.0)(5.0)^2 - f(1.6) = (12.0)(9.80)(0.80)$

Solve: $150\,\mathrm{J} - 1.6f = 94.1\,\mathrm{J}$, so $f = 55.9\,\mathrm{J}/1.6\,\mathrm{m}$ $\boxed{f = 35\,\mathrm{N}}$

EXECUTE: (b) states 1 to 3

Round trip, path length $2d = 3.2\,\mathrm{m}$: $K_1 + 0 - f(2d) = K_3 + 0$

$K_3 = 150\,\mathrm{J} - (34.95\,\mathrm{N})(3.2\,\mathrm{m}) = 38.2\,\mathrm{J}$, so $v_3 = \sqrt{2(38.2)/12.0}\,\mathrm{m/s}$

$\boxed{v_3 = 2.5\,\mathrm{m/s}}$

EVALUATE

$v_3 < v_1$, as predicted. Newton check down the ramp:

$a = g\sin 30° - f/m = 1.99\,\mathrm{m/s^2}$, $v_3 = \sqrt{2(1.99)(1.6)} = 2.5\,\mathrm{m/s}$. ✓

For each force: a potential-energy term, or part of W_{other}? Give the sign if it is in W_{other}.

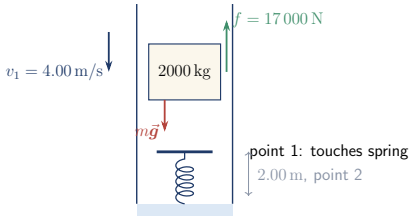
1. The weight of a crate sliding up a rough ramp.
2. The force of a spring that pushes the crate up the ramp.
3. Kinetic friction between the crate and the ramp.
4. The normal force of the ramp on the crate.
5. A trader pushing the crate up the ramp at constant speed.

Predict first

Sort all five before the answers appear.

Answers

(1) U_{grav}. (2) U_{el}. (3) $W_{\text{other}} < 0$. (4) $W_{\text{other}} = 0$: perpendicular to the motion. (5) $W_{\text{other}} > 0$.
Common error: counting the weight twice, once in U_{grav} and again as a work term.



Example 8: a design problem

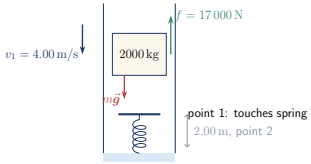
In a worst-case scenario, the cables of a $2000\,\mathrm{kg}$ lift in a tall building break. The lift is falling at $4.00\,\mathrm{m/s}$ when it first touches a cushioning spring at the bottom of the shaft. The spring is to stop it while compressing $2.00\,\mathrm{m}$. During this a safety clamp applies a constant $17\,000\,\mathrm{N}$ friction force. As the design consultant, what force constant k should the spring have?

Predict first

Is k larger or smaller than it would need to be with no clamp? Which terms are in U, which in W_{other}?

Example 8: a lift with broken cables, solution

CORE



IDENTIFY

Given: $m = 2000\,\mathrm{kg}$, $v_1 = 4.00\,\mathrm{m/s}$, $f = 17\,000\,\mathrm{N}$, compression $x = 2.00\,\mathrm{m}$, $v_2 = 0$. Find: k.

SET UP

Reference $y = 0$ at point 1 (spring at natural length): $y_1 = 0$, $y_2 = -2.00\,\mathrm{m}$. Gravity $\rightarrow U_{\text{grav}}$, spring $\rightarrow U_{\text{el}}$, clamp $\rightarrow W_{\text{other}}$. Use (7).

EXECUTE

Equation: $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$, so $\frac{1}{2}mv_1^2 + 0 - fx = 0 + mgy_2 + \frac{1}{2}kx^2$

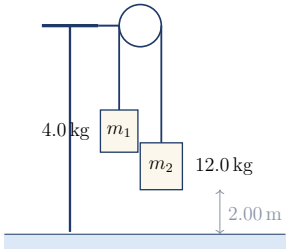
Terms: $K_1 = \frac{1}{2}(2000)(4.00)^2 = 16\,000\,\mathrm{J}$;
$W_{\text{other}} = -(17\,000)(2.00) = -34\,000\,\mathrm{J}$;
$mgy_2 = (2000)(9.80)(-2.00) = -39\,200\,\mathrm{J}$

Solve: $\frac{1}{2}kx^2 = 16\,000 - 34\,000 + 39\,200 = 21\,200\,\mathrm{J}$, so

$k = 2(21\,200)/(2.00)^2$ $\quad \boxed{k = 1.06 \times 10^4\,\mathrm{N/m}}$

EVALUATE

With no clamp, k would be $2(16\,000 + 39\,200)/4 = 2.76 \times 10^4\,\mathrm{N/m}$: larger, so smaller with the clamp, as predicted. Check: at the bottom the upward forces are $21\,200 + 17\,000 = 38\,200\,\mathrm{N}$ against a $19\,600\,\mathrm{N}$ weight, so $a = 9.3\,\mathrm{m/s^2} \approx 1\,g$. ✓



Key idea

The tensions do work of equal size and opposite sign: they cancel.

Example 9

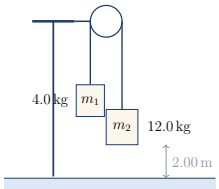
At a building site a $12.0\,\mathrm{kg}$ sack and a $4.00\,\mathrm{kg}$ sack hang from a light rope over a frictionless pulley. The $12.0\,\mathrm{kg}$ sack is released from rest $2.00\,\mathrm{m}$ above the ground. (a) Find its speed just before it lands, using energy. (b) Find it again using Newton's second law and kinematics. (c) Which method would you use if I asked for the time, or the tension?

Predict first

Is the speed more or less than the $6.26\,\mathrm{m/s}$ of a single sack falling freely $2.00\,\mathrm{m}$?

Example 9: energy or Newton? Two sacks over a pulley, solution

CORE



Key idea

The tensions do work of equal size and opposite sign: they cancel.

IDENTIFY

Given: $m_1 = 12.0\,\text{kg}$, $m_2 = 4.00\,\text{kg}$, $h = 2.00\,\text{m}$, both start at rest.
Find: v just before m_1 lands.

(a) Energy, system of both sacks and the Earth

The rope is inextensible, so both sacks move h and have speed v. The tension does $-Th$ on one and $+Th$ on the other: they cancel. Only gravity does net work, so use (4).

Equation: $K_1 + U_1 = K_2 + U_2$, so $0 = \frac{1}{2}(m_1 + m_2)v^2 - (m_1 - m_2)gh$

Substitute: $(12.0 - 4.00)(9.80)(2.00) = \frac{1}{2}(16.0)v^2$, so $156.8 = 8.00v^2$

$\boxed{v = 4.43\,\text{m/s}}$

(b) Newton (Week 3) and kinematics (Week 2)

$m_1 g - T = m_1 a$ and $T - m_2 g = m_2 a$. Add:

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} = \frac{(8.00)(9.80)}{16.0} = 4.90\,\text{m/s}^2$$

$v^2 = 2ah = 19.6$, so $\boxed{v = 4.43\,\text{m/s}}$; also $T = m_2(g + a) = 58.8\,\text{N}$,

$t = \sqrt{2h/a} = 0.90\,\text{s}$

(c) and EVALUATE

Below $6.26\,\text{m/s}$, as predicted. Both agree. For the time or tension only Newton works; for the speed alone, energy is shorter. ✓

A new question: which forces can be stored?

So far

Gravity and the spring have a potential energy: $W = -\Delta U$. Friction has none: its work enters as W_{other}.

Not yet answered

What separates a force with a potential energy from one without? Given $U(x)$, what is the force, and where can the particle go?

Opening question

Two climbers go from the base of Mount Afadjato to the summit, one by a steep path, the other by a long, gentle one. Gravity does the same work on both. Does friction? Does *every* force have a potential energy? How would we test?

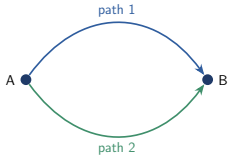
eg. Mount Afadjato, the highest mountain in Ghana, in the Volta Region.

Predict first

Recall the stones, Kofi's ramp and the tomato crate. For which forces did the path not matter?

Rule

All forces are either conservative or nonconservative.



Key idea

If $W_1 = W_2$ for any two paths A to B, then going out by path 1 and back by path 2 reverses one work: the total around the loop is zero.

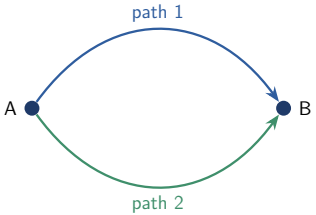
Definition: conservative force

A force that allows two-way conversion between kinetic and potential energy is a **conservative force**. The work it does has four properties:

1. it can be written as the difference between the initial and final values of a potential-energy function;
2. it is reversible;
3. it is independent of the path and depends only on the starting and ending points;
4. for a closed path, with the same start and end, the total work is zero.

Define. A *closed path* returns to its start. $W_{A\to B}$ is the work of the force along a chosen route (J).

Explain. Properties 3 and 4 are the same statement. Gravity and the ideal spring pass both tests. When only conservative forces do work, $E = K + U$ is constant, Equation (4).



Key idea

Go out by path 1 and come back by path 2. Reversing a path reverses the sign of the work of a conservative force.

Rule: the closed-loop test

A force is conservative if and only if the work it does round every closed path is zero.

Define. W_1 and W_2 are the works of the force from A to B along path 1 and along path 2 (J). W_{loop} is the work round the loop A→B (path 1) then B→A (back along path 2).

Explain.
1. Work is $\int \vec{F} \cdot \mathrm{d}\vec{r}$. Going back along path 2 reverses every $\mathrm{d}\vec{r}$, so the work from B to A is $-W_2$.
2. Therefore $W_{\text{loop}} = W_1 + (-W_2) = W_1 - W_2$.
3. If the work is the same on every path, $W_1 = W_2$ and $W_{\text{loop}} = 0$. Conversely, if $W_{\text{loop}} = 0$ for every loop, then $W_1 = W_2$ for any two paths. The two statements are the same test.

Friction fails it. Friction always opposes the motion, so going back it again does negative work: $W_{\text{loop}} = -f_k(d_1 + d_2) \neq 0$.

Opening question, answered. Friction: no. When the motion reverses, so does the friction force: negative work both ways, and the longer path costs more. **Not every force has a potential energy.**

Definition: nonconservative force

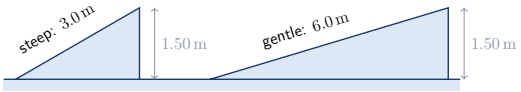
A force that is not conservative is **nonconservative**: its work cannot be written as a potential-energy function. Kinetic friction and fluid drag lose mechanical energy: **dissipative forces**. Others add it: in an exploding firecracker the chemical reaction of gunpowder with oxygen turns chemical energy into the large kinetic energy of the fragments, and the process is not reversible. A push or a motor also adds energy, through W_{other}.

• METHOD – Testing a force

1. TWO PATHS: pick two paths between the same two points. Find the work of the force on each.
2. COMPARE: the same on every path means conservative; different means nonconservative.
3. LOOP: or find the work around a closed path: zero means conservative.
4. ASSIGN: conservative forces go into U; nonconservative forces go into W_{other}.

WATCH OUT: a force that does zero work in one case is not thereby conservative (the normal force on a level floor).
CHECK: does the work of friction grow with the path length?

Example 10: a sack of maize up two ramps



Key idea

Same start, same platform, two routes: compare the works of gravity and of friction.

Example 10

A trader pushes a $10.0\,\mathrm{kg}$ sack of maize at constant speed up either a steep ramp ($3.0\,\mathrm{m}$ long) or a gentle ramp ($6.0\,\mathrm{m}$ long) to a platform $1.50\,\mathrm{m}$ high. The push is parallel to the ramp and $\mu_k = 0.20$. On each ramp, find the work of gravity, the work of friction and the work she does.

Predict first

Which ramp needs the larger force? Which needs the larger *work*? Is the work of gravity the same?

eg. maize and cement sacks loaded onto lorries and platforms at the markets.

Example 10: a sack of maize up two ramps, solution

CORE

IDENTIFY

Given: $m = 10.0\,\text{kg}$, rise $h = 1.50\,\text{m}$ on both, lengths $3.0\,\text{m}$ and $6.0\,\text{m}$, $\mu_k = 0.20$, constant speed so $\Delta K = 0$. Find: W_{grav}, W_f and W_{push} on each.

SET UP

(1) for gravity; $f_k = \mu_k n = \mu_k mg\cos\theta$ with $\sin\theta = h/L$; $W_f = -f_k L$ (8a). The theorem, $W_{\text{push}} + W_{\text{grav}} + W_f = 0$. *Reuse:* Lecture 4 (5), (6)

EXECUTE: steep ramp ($\sin\theta = 0.500$, $\cos\theta = 0.866$)

Gravity:
$W_{\text{grav}} = -mgh = -(10.0)(9.80)(1.50) = -147\,\text{J}$
Friction force: $f_k = 0.20(98.0\,\text{N})(0.866) = 17.0\,\text{N}$
Friction work: $W_f = -(17.0\,\text{N})(3.0\,\text{m}) = -50.9\,\text{J}$
Push: $W_{\text{push}} = 147 + 50.9$ $\boxed{W_{\text{push}} = 198\,\text{J}}$

EXECUTE: gentle ramp ($\sin\theta = 0.250$, $\cos\theta = 0.968$)

Gravity: $W_{\text{grav}} = -147\,\text{J}$, the same
Friction force: $f_k = 0.20(98.0\,\text{N})(0.968) = 19.0\,\text{N}$
Friction work: $W_f = -(19.0\,\text{N})(6.0\,\text{m}) = -114\,\text{J}$
Push: $W_{\text{push}} = 147 + 114$ $\boxed{W_{\text{push}} = 261\,\text{J}}$

EVALUATE

Gravity: the same on both, as for any conservative force. Friction: not the same, so it has no potential energy. The gentle ramp needs the smaller force ($F = mg\sin\theta + f_k$: $66.0\,\text{N}$ steep, $43.5\,\text{N}$ gentle) but more work, because friction acts over twice the distance. Check: $(66.0)(3.0) = 198\,\text{J}$ and $(43.5)(6.0) = 261\,\text{J}$. ✓

But what if... friction took energy out of the mechanical bars on each ramp? The next rule says where it goes.

Rule: law of conservation of energy

$$\Delta K + \Delta U + \Delta U_{\text{int}} = 0 \qquad (9)$$

Define. ΔK, ΔU and ΔU_{int} are the changes (J) in the kinetic, potential and internal energy of the **system**, between two states. The system is isolated: no outside force does work on it.

Explain. Energy is never created or destroyed; it changes form or moves between parts of the system. With kinetic friction, $\Delta U_{\text{int}} = -W_{\text{other}} = f_k d$ from (8b), which turns (9) into (7). If an outside force does work, include the agent in the system, or count its work as W_{other}.

Valid for every process we know.

Concept check

Identify the energy changes.
(a) A trotro brakes to rest on the Accra–Kumasi road.
(b) A kayayo climbs market steps at constant speed.
(c) Water falls through the turbines at Akosombo.

Predict first

For each, name the energy that decreases and the energy that increases.

(a) $K \to U_{\text{int}}$: brakes and tyres get hot. (b) Her chemical (internal) energy decreases; U_{grav} and the warmth of her muscles increase. (c) $U_{\text{grav}} \to K \to$ electrical energy, and a little warming.

Rule: force from potential energy

$$F_x(x) = -\frac{\mathrm{d}U(x)}{\mathrm{d}x} \tag{10}$$

Define. $U(x)$ is the potential-energy function (J) of a conservative force, in one dimension. F_x is the x-component of that force (N). $\mathrm{d}U/\mathrm{d}x$ is the slope of the U–x graph (J/m = N).

Explain. A small step $\mathrm{d}x$ does work $F_x\,\mathrm{d}x = -\mathrm{d}U$, by (2). So the force is the *negative slope* of U: it pushes toward **lower** U, and is zero where the slope is zero. Check: $U = \frac{1}{2}kx^2$ gives $F_x = -kx$, Hooke's law. $U = mgy$ gives $F_y = -mg$, downward.

Valid for conservative forces. In three dimensions each component is $-\partial U/\partial x$, $-\partial U/\partial y$, $-\partial U/\partial z$ (extension, next frame).

Predict first

On a graph of $U(x)$ the curve rises steeply to the right of a point. Predict the direction and the relative size of the force there.

Steep slope, positive: a large force toward $-x$, down the slope.

Common error

The *sign of* U does not give the direction of the force. The sign of the *slope* does. You can add a constant to U without changing any force.

Extension: the force from U in three dimensions

Rule: force from $U(x, y, z)$

$$\vec{\boldsymbol{F}} = -\left(\frac{\partial U}{\partial x}\,\hat{\boldsymbol{\imath}} + \frac{\partial U}{\partial y}\,\hat{\boldsymbol{\jmath}} + \frac{\partial U}{\partial z}\,\hat{\boldsymbol{k}}\right) = -\nabla U \qquad (10')$$

Define. $\partial U/\partial x$ is the *partial derivative*: differentiate with respect to x and hold y and z fixed (J/m = N). Each component is one-dimensional Equation (10) for that coordinate. ∇U ("grad U") is the vector of the three.

Explain. The force points toward the steepest decrease of U, perpendicular to surfaces of constant U. Checks:
$U = mgz \Rightarrow \vec{\boldsymbol{F}} = -mg\,\hat{\boldsymbol{k}}$.
$U = \frac{1}{2}k(x^2 + y^2) \Rightarrow \vec{\boldsymbol{F}} = -k(x\,\hat{\boldsymbol{\imath}} + y\,\hat{\boldsymbol{\jmath}})$, a spring force in a plane. Not assessed.

Example

A particle has $U = 3.0\,x^2 y$ (J, x and y in m). Find the force at $(1.0\,\mathrm{m},\ 2.0\,\mathrm{m})$.

Predict first

U grows with x and with y there. Predict the signs of F_x and F_y.

Example (solution). A particle has $U = 3.0\,x^2 y$ (J, x and y in m). Find the force at $(1.0\,\mathrm{m},\ 2.0\,\mathrm{m})$.

IDENTIFY

Given $U = 3.0x^2y$; point $(1.0, 2.0)$. Find $\vec{\boldsymbol{F}}$.

SET UP

Use (10′): differentiate with the other variable fixed.

EXECUTE

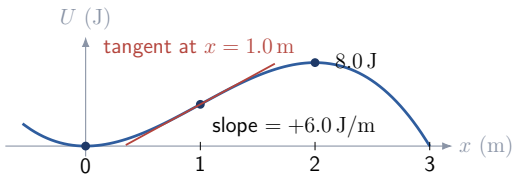
$F_x = -\partial U/\partial x = -6.0xy = -6.0(1.0)(2.0) = -12.0\,\mathrm{N}$

$F_y = -\partial U/\partial y = -3.0x^2 = -3.0(1.0)^2 = -3.0\,\mathrm{N}$

$$\boxed{\vec{\boldsymbol{F}} = (-12.0\,\mathrm{N})\,\hat{\boldsymbol{\imath}} + (-3.0\,\mathrm{N})\,\hat{\boldsymbol{\jmath}}}, \quad |\vec{\boldsymbol{F}}| = \sqrt{144 + 9} = 12.4\,\mathrm{N}$$

EVALUATE

Both negative, as predicted: the force points toward lower U. ✓



Key idea

The tangent at $x = 1.0\,\mathrm{m}$ slopes upward, so the force points to the left.

Example 11

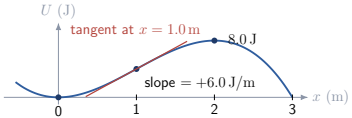
A particle moves along the x-axis with potential energy $U(x) = 6.0x^2 - 2.0x^3$ (U in joules, x in metres). Find (a) the force at $x = 1.0\,\mathrm{m}$, and (b) the equilibrium positions.

Predict first

From the graph, does the force at $x = 1.0\,\mathrm{m}$ point toward $+x$ or $-x$? Where is the slope zero?

Example 11: from a formula for U to the force, solution

CORE



Key idea

A minimum of U is stable, a maximum is unstable.

IDENTIFY

Given: $U(x) = 6.0x^2 - 2.0x^3$ (J, with x in m). Find: F_x at $x = 1.0\,\mathrm{m}$; the positions where $F_x = 0$.

SET UP

Use (10), $F_x = -\mathrm{d}U/\mathrm{d}x$, with $\mathrm{d}(x^n)/\mathrm{d}x = nx^{n-1}$. *Reuse:* Lecture 4 (7), the reverse step

EXECUTE (a)

Equation: $F_x = -\dfrac{\mathrm{d}U}{\mathrm{d}x} = -(12x - 6x^2) = -12x + 6x^2$

Substitute $x = 1.0$: $F_x = -12(1.0) + 6(1.0)^2$ $\boxed{F_x = -6.0\,\mathrm{N}}$

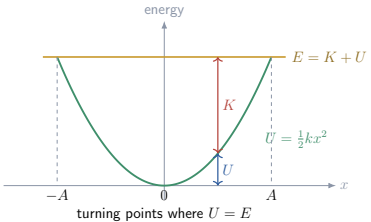
EXECUTE (b)

$F_x = 0$: $6x(x - 2) = 0$, so $\boxed{x = 0 \text{ and } x = 2.0\,\mathrm{m}}$

EVALUATE

Slope $+6.0\,\mathrm{J/m}$, force $-6.0\,\mathrm{N}$, toward lower U, as predicted. $x = 0$ is a minimum of U: **stable**. $x = 2.0\,\mathrm{m}$ is a maximum: **unstable**. ✓

Reading an energy diagram



turning points where $U = E$

Key idea

The gap between the E line and the U curve is K.

Predict first

$E = 2.0\,\mathrm{J}$, $U = 0.5\,\mathrm{J}$ at some x, $m = 0.50\,\mathrm{kg}$. Predict K there, and whether v is more or less than $2\,\mathrm{m/s}$.

Rule: the energy diagram

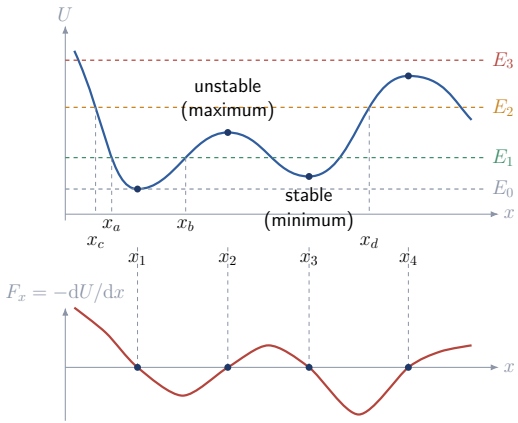
$$K = E - U \geq 0 \qquad \text{(11a)}$$

$$v = \sqrt{2(E-U)/m} \qquad \text{(11b)}$$

Define. An **energy diagram** is a graph of $U(x)$ with the constant total energy E drawn as a horizontal line. K is the gap from the curve up to the line (J).

Explain. A glider on an air track has $U = \frac{1}{2}kx^2$. It can be only where $E \geq U$, because $K < 0$ is impossible. **Turning points** are where $U = E$, so $K = 0$: the glider stops and reverses ($x = \pm A$). It is fastest where U is lowest ($x = 0$).

Valid for one-dimensional motion with only conservative forces doing work.



Explain. The lower graph is $F_x = -\mathrm{d}U/\mathrm{d}x$, from (10), for the curve above it.

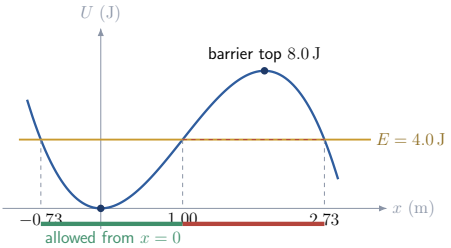
Equilibrium where the slope is zero ($F_x = 0$). A **minimum** (x_1, x_3) is stable: a small displacement meets a force back toward it. A **maximum** (x_2, x_4) is unstable.

E_0**:** the smallest possible energy; the particle rests at x_1.
E_1**:** trapped between x_a and x_b.
E_2**:** trapped between x_c and x_d; it cannot cross the hill at x_4.
E_3**:** above every barrier on the right: the particle can escape to large x.

Common error

A region where $E \geq U$ is allowed, but it is reachable only if no $U > E$ lies between it and the start.

Example 12: trapped or not?



Key idea

Sketch U and the line $E = 4.0\,\text{J}$ first. Turning points are where they cross.

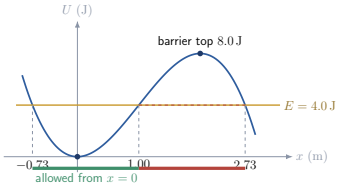
Example 12

For $U(x) = 6.0x^2 - 2.0x^3$ (J, x in m), a $0.50\,\text{kg}$ particle at $x = 0$ has total energy $E = 4.0\,\text{J}$. Find (a) the turning points of its motion, (b) its speed at $x = 0$, and (c) the energy above which it can escape to large x.

Predict first

Which way can it go from $x = 0$ before it turns? Is it fastest at $x = 0$? Could it reach $x = 3\,\text{m}$?

Example 12: trapped or not? Solution



Key idea

The region $x > 2.73\,\text{m}$ has $E > U$ but is cut off by the barrier.

IDENTIFY

Given: $U = 6.0x^2 - 2.0x^3$, $m = 0.50\,\text{kg}$, $E = 4.0\,\text{J}$ at $x = 0$. Find: turning points; v at $x = 0$; the escape energy.

SET UP

Use (11): $K = E - U \geq 0$; turning points where $U = E$.

EXECUTE (a)

Equation: $6x^2 - 2x^3 = 4$, so
$x^3 - 3x^2 + 2 = (x-1)(x^2 - 2x - 2) = 0$
Roots: $x = 1.00$ and $x = 1 \pm \sqrt{3}$, that is $2.73\,\text{m}$ and $-0.73\,\text{m}$
From $x = 0$, $U < E$ between -0.73 and $1.00\,\text{m}$:
$\boxed{\text{turning points } x = -0.73\,\text{m and } +1.00\,\text{m}}$

EXECUTE (b) and (c)

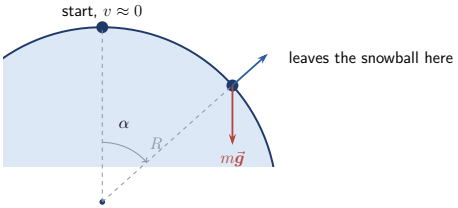
(b) $K = E - U(0) = 4.0\,\text{J} = \frac{1}{2}(0.50)v^2$, so $v = \sqrt{2(4.0)/0.50}$
$\boxed{v = 4.0\,\text{m/s}}$

(c) The barrier top is $U(2.0) = 8.0\,\text{J}$. Escape needs $\boxed{E > 8.0\,\text{J}}$.

EVALUATE

v is greatest at $x = 0$, the minimum of U, as predicted. The region $x > 2.73\,\text{m}$ is allowed ($E > U$) but separated from the start by $U > E$ between 1.00 and $2.73\,\text{m}$: it cannot be reached. ✓

Your turn: leaving the snowball



Key idea

Contact is lost when the normal force reaches zero. Energy gives v; Newton at that point gives n.

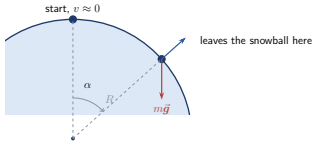
Independent problem

A skier starts at the top of a very large frictionless snowball with a very small speed and skis straight down its side. At what angle α between the vertical and her radius does she lose contact with the snow?

Predict first

Near the top she presses hard on the snow; lower down the snow pushes less. Between $0°$ and $90°$: nearer $20°$, $50°$ or $80°$?

Your turn: leaving the snowball, solution



Key idea

Contact is lost when the normal force reaches zero. Energy gives v; Newton at that point gives n.

IDENTIFY

Given: radius R, $v_1 \approx 0$ at the top, no friction. Find: α at which $n = 0$.

SET UP

n does no work: energy for v (4). Newton toward the centre for n.
Reference $y = 0$ at the centre: $y_1 = R$, $y_2 = R\cos\alpha$.

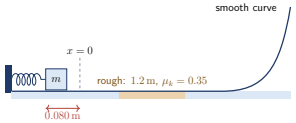
EXECUTE

Energy: $mgR = \frac{1}{2}mv^2 + mgR\cos\alpha$, so $v^2 = 2gR(1 - \cos\alpha)$
Newton toward the centre: $mg\cos\alpha - n = mv^2/R$
Leaves when $n = 0$: $g\cos\alpha = 2g(1 - \cos\alpha)$, so $3\cos\alpha = 2$

$$\boxed{\cos\alpha = \tfrac{2}{3}, \quad \alpha = 48.2^\circ}$$

EVALUATE

Near 50°, as predicted. It does not depend on m or R. Check: $n = mg(3\cos\alpha - 2)$ is mg at the top and falls to zero at $\cos\alpha = 2/3$. ✓



Application problem: no method is named

A $0.40\,\mathrm{kg}$ block is pushed against a horizontal spring ($k = 1200\,\mathrm{N/m}$), compressing it $0.080\,\mathrm{m}$, and released from rest. The block is not attached. The track is smooth except for a rough strip $1.2\,\mathrm{m}$ long ($\mu_k = 0.35$), followed by a smooth curve.

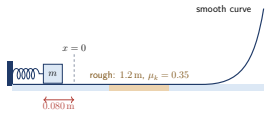
(a) How high up the curve does the block rise?

(b) The block slides back. How far does it compress the spring when it returns?

Predict first

Sketch the energy bars. Does the block come back with less energy than it started with? Which tool does each stage need?

Integrated problem: solution (1)



Key idea

Spring and gravity go into U.
Friction on the strip is W_{other}.

IDENTIFY

Given: $m = 0.40\,\text{kg}$, $k = 1200\,\text{N/m}$, $x_1 = 0.080\,\text{m}$, strip $1.2\,\text{m}$, $\mu_k = 0.35$, $v_1 = 0$. Find: (a) the greatest height H; (b) the compression when it returns.

SET UP

Use (7). Reference $y = 0$ at the track. Friction acts only on the strip, once each way; the curve is smooth. *Reuse:* (6) spring, (8a) friction, (3) gravity

EXECUTE: the common pieces

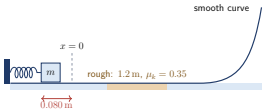
Spring: $U_{\text{el}} = \frac{1}{2}kx_1^2 = \frac{1}{2}(1200\,\text{N/m})(0.080\,\text{m})^2 = 3.84\,\text{J}$
Friction force: $f_k = \mu_k mg = (0.35)(0.40)(9.80) = 1.372\,\text{N}$
Per crossing: $W_f = -(1.372\,\text{N})(1.2\,\text{m}) = -1.65\,\text{J}$

EXECUTE: (a) release to the highest point ($K = 0$)

Equation: $U_{\text{el},1} + W_f = mgH$, so $3.84 - 1.65 = (0.40)(9.80)H$

$$H = \frac{2.19\,\text{J}}{3.92\,\text{N}} = 0.56\,\text{m}$$

Integrated problem: solution (2)



Key idea

Each crossing removes $1.65\,\text{J}$. After the round trip, less than the full $3.84\,\text{J}$ remains.

EXECUTE: (b) release to maximum compression on return ($K = 0$)

Two crossings: $W_f = -2(1.65\,\text{J}) = -3.29\,\text{J}$

Equation: $U_{\text{el},1} + W_f = \frac{1}{2}kx^2$, so $3.84 - 3.29 = 0.547\,\text{J} = \frac{1}{2}(1200)x^2$

$$x = \sqrt{\frac{2(0.547\,\text{J})}{1200\,\text{N/m}}} \qquad \boxed{x = 0.030\,\text{m}}$$

EVALUATE

$3.0\,\text{cm} < 8.0\,\text{cm}$: it returns with less energy, as predicted. Bookkeeping with (9): $\Delta U_{\text{int}} = 3.29\,\text{J}$, and $3.29 + 0.547 = 3.84\,\text{J}$. ✓

MODIFY

If μ_k were 0.70, one crossing costs $3.29\,\text{J}$, leaving $0.55\,\text{J}$: the block rises $0.14\,\text{m}$, comes back, and stops $0.20\,\text{m}$ into the strip. It never returns to the spring.

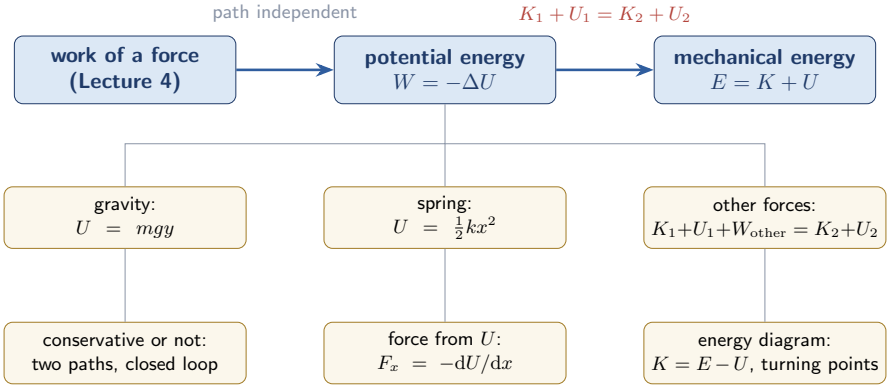
Which tools, and why?

Spring and gravity went into U because their work is path independent; friction went into W_{other} because it is not. One equation, (7), covered every stage.

Common mistakes and the correct habit

Mistake	Instead
Taking the sign of U as the direction of the force.	The force follows the slope: $F_x = -\mathrm{d}U/\mathrm{d}x$.
Calling a force conservative because it did no work in one case.	Test with two paths or a closed loop.
Saying every region with $E \geq U$ is reachable.	It must connect to the start without crossing $U > E$.
Leaving out the internal-energy bar.	Mechanical energy lost to friction reappears as U_{int}.
Forgetting that a maximum of U is an unstable equilibrium.	A small displacement meets a force that pushes it further away.

But what if... you can now sort forces, find a force from U and read an energy diagram? Pull the whole week together: the Summary.



law of conservation of energy: $\Delta K + \Delta U + \Delta U_{\text{int}} = 0$

Work that does not depend on the path can be stored as potential energy. With only stored forces working, $E = K + U$ is constant. Other forces change E by their work.

Key ideas

• **The work of gravity depends only on the heights:** $W_{\text{grav}} = mg(y_1 - y_2) = -\Delta U_{\text{grav}}$**, with** $U_{\text{grav}} = mgy$**.**
• Only ΔU is physical. The reference level is a choice, and U can be negative. U belongs to the system, not to one object.
• **If only conservative forces do work,** $K_1 + U_1 = K_2 + U_2$**. The sum** $E = K + U$ **is conserved;** K **alone is not.**
• Energy gives speeds, heights and compressions. Use Newton's second law for a force at a point, an acceleration or a time.
• **A spring stores** $U_{\text{el}} = \frac{1}{2}kx^2$**, measured from the natural length; it is never negative.**
• **Other forces change** E **by their work:** $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$**. Friction gives** $W_f = -f_k d$ **with the path length.**
• A conservative force has path-independent work and zero work around a closed loop. Friction fails both tests.
• **Energy is never created or destroyed:** $\Delta K + \Delta U + \Delta U_{\text{int}} = 0$**. Friction turns mechanical energy into internal energy.**
• The force is the negative slope of U: $F_x = -\mathrm{d}U/\mathrm{d}x$. On an energy diagram $K = E - U \geq 0$, turning points have $K = 0$, minima are stable, maxima unstable.
• **Habits:** sketch, predict, calculate, check units, sign and a second method.

Reference tools: the next slides (equation map, master toolkit). The Reference deck holds the seven toolkits and the equation-anatomy slides.

Reference tools: the equation map

Idea	Equation	Conditions and units
(1) Work of gravity	$W_{\text{grav}} = mg(y_1 - y_2)$	near the surface; any path; J
(2), (3) Gravitational U	$W_{\text{grav}} = -\Delta U_{\text{grav}}, \quad U_{\text{grav}} = mgy$	y up from a chosen level; J
(4) Conservation	$K_1 + U_1 = K_2 + U_2$	only conservative forces do work
(5) Work of a spring	$W_{\text{el}} = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2$	x from natural length; J
(6) Elastic U	$W_{\text{el}} = -\Delta U_{\text{el}}, \quad U_{\text{el}} = \frac{1}{2}kx^2$	Hooke's law range; ≥ 0
(7) Other forces	$K_1 + U_1 + W_{\text{other}} = K_2 + U_2, \quad U = U_{\text{grav}} + U_{\text{el}}$	any forces, once sorted
(8a), (8b) Friction	$W_f = -f_k d, \quad \Delta U_{\text{int}} = +f_k d$	d path length; sliding
(9) Energy law	$\Delta K + \Delta U + \Delta U_{\text{int}} = 0$	isolated system
(10) Force from U	$F_x = -\mathrm{d}U/\mathrm{d}x$	conservative; J/m = N
(11a), (11b) Energy diagram	$K = E - U \geq 0, \quad v = \sqrt{2(E-U)/m}$	turning point $K = 0$

• MASTER TOOLKIT – Energy

1. IDENTIFY The object, and what is asked: a speed, a height, a distance, a compression, a force or a time.
2. STATES Two positions, with heights, speeds and spring extensions. One reference level.
3. SORT THE FORCES Into U_{grav}, U_{el} or W_{other}. Perpendicular forces give zero.
4. PREDICT Does E stay constant, fall or rise? Rough size of the answer.
5. METHOD Energy, or Newton's second law.
6. EXECUTE $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$; or $F_x = -\mathrm{d}U/\mathrm{d}x$ and $K = E - U$.
7. EVALUATE Units, signs, faster than the frictionless case? A second method.

Step 3: the key move for each force

Gravity: $U_{\text{grav}} = mgy$, from the height change only.
Spring: $U_{\text{el}} = \frac{1}{2}kx^2$, x from the natural length.
Friction: $W_f = -f_k d$, path length; $\Delta U_{\text{int}} = +f_k d$.
Push or motor: $W = Fs\cos\phi$, positive.
Normal force, tension: $W = 0$ if perpendicular to the motion.
Unknown force: two-path test, then $F_x = -\mathrm{d}U/\mathrm{d}x$.

Framework: IDENTIFY (1), SET UP (2–5), EXECUTE (6), EVALUATE (7).

Where this goes next

Lecture 6: gravitation

$U = mgy$ is the small-height form of $U = -GMm/r$. Escape speed is an energy-diagram question: how large must E be to pass every barrier?

Lecture 7: momentum

Energy and momentum are the two conservation laws for collisions. Energy tells you what is kept as motion; momentum gives the directions.

Lectures 8 and 11: rotation, heat

Rotational kinetic energy joins K in E. The internal-energy bar becomes temperature and heat, and (9) becomes the first law of thermodynamics.

Earlier weeks, now working together

Lecture 4's theorem $W_{\text{tot}} = \Delta K$ gives (4) and (7). Week 1's dot product gives the work of gravity. Week 3's free-body diagram sorts the forces and gives the force at a point.

Physical meaning

Energy is the central bookkeeping idea of the course. Energy diagrams describe molecular bonds and oscillations later on, using exactly the reading skills from this week.

Checklist: can I . . . CORE

- ☐ calculate $U_{\text{grav}} = mgy$ with a stated reference level?
- ☐ explain why only ΔU matters?
- ☐ state when $K_1 + U_1 = K_2 + U_2$ holds?
- ☐ use it on straight and curved paths, and use Newton for a force?
- ☐ calculate $U_{\text{el}} = \frac{1}{2}kx^2$ from the natural length?
- ☐ sort the forces on a diagram into U and W_{other}?
- ☐ use the energy equation with friction, and say where the energy goes?
- ☐ test whether a force is conservative?
- ☐ state the law of conservation of energy?
- ☐ find $F_x = -\mathrm{d}U/\mathrm{d}x$?
- ☐ read turning points, K and stability from an energy diagram?
- ☐ select the appropriate model in an unfamiliar problem?

Next

Tutorial sheet: extra gravitational, spring, friction, force-from-potential and energy-diagram problems, with a ladder from independent practice to exam-style items. The spring and rough ramp, the loop-the-loop and the double well, moved out of the lectures, are solved in section D. The Reference deck holds the material cut from the live lecture. The supplementary notes hold the derivative refresher, the closed-loop proof, the three-dimensional gradient, the vertical-loop minimum speed, the stability test, and the preview of $U = -GMm/r$.

Hope Donglo

Room 24, Department of Physics

hdonglo@ug.edu.gh

Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html